

Solution Requirements

Date	1 October 2026
Team ID	6ab501466f290be8a6b4f5d3
Project Name	ComicCraft – AI Comic Story Creator
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No user login is needed. The Gemini API key is loaded securely from a .env file (python-dotenv) and is never shown in the interface.	Medium
2	Authorization levels	One user role: the app user, who can enter a story idea, generate a comic and download the PDF. The API key is accessible only to the backend code.	Low
3	External interfaces	Google Gemini text model (story and panel script) and Gemini image model (panel pictures), accessed through the google-genai library.	High
4	Transactions processing	Each request flows from the user's story idea to a story script, panel prompts, panel images, page layout and PDF. Data is held only for the user's session; no payments are involved.	High
5	Reporting	On-screen preview of the generated story and panels, and a downloadable comic PDF created with FPDF2.	High
6	Business rules, etc.	The story is split into a fixed number of panels, each with an image and a caption. The comic follows the chosen genre and art style, and an empty story idea is not accepted.	Medium
7	Compliance to laws or regulations	Follow Google Gemini API terms of use, keep the API key private, and do not collect or store personal data of users.	Medium
8	Other	Users can choose the genre and art style, and regenerate a panel they are not happy with.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The story script and panel images should be generated within a reasonable waiting time, with a progress message shown to the user.	Story script in under 15 seconds; each panel image in under 30 seconds (depends on the Gemini API)
2	Scalability	Each user session runs separately in Streamlit, and the code is split into modules so new features can be added easily.	Several users can use the app at the same time during a demo
3	Security & Data Privacy	The API key is stored in a .env file kept out of the GitHub repository, and no personal user data is stored.	No API key in the source code or repository
4	Reliability & Availability	If the Gemini API fails or reaches its limit, the app shows a clear error message and lets the user try again.	App does not crash on API errors; retry is available
5	Usability & Accessibility	A simple interface with a prompt box, genre and style selectors and a generate button, with readable text and clear labels.	A comic can be made in 3 steps: enter idea, choose style, click generate
6	Other	Maintainability and portability: separate modules for the app, story, images, layout and export, running on any system with Python and internet access.	5 separate modules; runs after installing the listed libraries